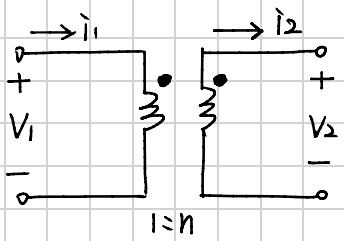
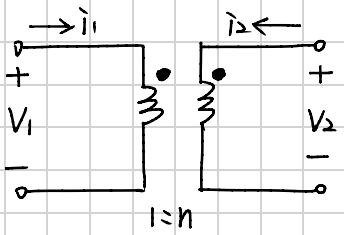
理想变压器的电压和电流关系



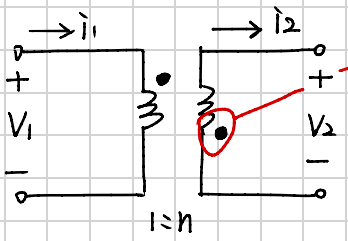
$i_1 = n i_2$　　$V_2 = n V_1$

电流一流进同名端，一流出同名端，电流关系为正



$i_1 = -n i_2$

电流同时流进同名端
电流关系为负



V_2、V_1 同名端反过来后，电压关系反转，

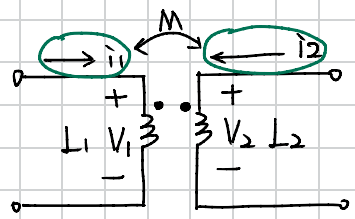
$V_2 = -nV_1$

$i_1 = -n i_2$

对于原线圈，i_1和v_1同向，改为$L_1\frac{di_1}{dt}$
对于副线圈，i_2和v_2同向，改为$M\frac{di_2}{dt}$
$v_1 = L_1\frac{di_1}{dt} + M\frac{di_2}{dt}$　反之

磁耦合：

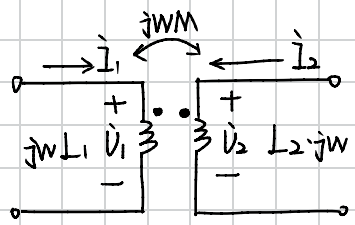
时域



$$V_1 = L_1\frac{di_1}{dt} + M\frac{di_2}{dt}$$

$$V_2 = L_2\frac{di_2}{dt} + M\frac{di_1}{dt}$$

相量域



$$\dot{V}_1 = jwL_1\dot{I}_1 + jwM\dot{I}_2$$

$$\dot{V}_2 = jwL_2\dot{I}_2 + jwM\dot{I}_1$$

$M = \sqrt{L_1 L_2} \cdot K$　K为耦合系数